

46 Improvements

Limitations	Improvement
Two (sets of) readings are not enough to draw a valid conclusion.	Take more readings and **plot a graph.**

General:

Limitations	Improvements
Difficulty judging the exact instant ./ threshold when a maximum occurs or when the motion starts, stops or changes direction	Use a [close-up] video **with [apparatus**] in view and play back **frame by frame** to determine the [value] more accurately Use a reference mark / scale / screen / objective indicator Change quantity by smaller increments Use an appropriate sensor to detect the event
Object does not have the ideal or uniform shape assumed in the calculation A dimension or physical property is not uniform throughout the object	Measure from two or more sides and average Use more rigid component or support (tension / clamp/ tape) the component so that it is rigid and shape is maintained.
A wire, rod, tube or similar component is not straight	

Limitations	Improvements
Mass, volume or dimensions of an additional component are not included in the calculation	Measure the extra mass/volume /geometry and include it in the calculation Use a lighter/negligible alternative attachment
The available adjustment or increment is too large for precise control	Use smaller/lighter increments
Required steady condition is not reached or maintained	Allow/check that the steady/terminal condition has been reached Reduce disturbances by increasing control of the property.
A physical effect or model assumption is not properly accounted for	Change the apparatus/material/model to remove the unaccounted effect

Measurement:

Limitations	Improvements
Measured value or change in value is small, giving a large percentage uncertainty	Use a more precise instrument; Increase the measured change/value instead
Parallax error when measuring a position, length or angle Part of the apparatus obstructs the known required measurement point Instrument resolution is insufficient for the quantity being measured Available instrument or apparatus range is insufficient for the required measurement/variation. Difficult to judge the exact fraction of a complete turn, scale division or distance.	Small objects: Use callipers / micrometer, instruments that clamp onto the exact *edges* OR travelling microscope Larger objects: Use a [marker / set square] to mark the object's position on the ruler. Clamp ruler / protractor / [component] (vertically / horizontally) Rotational: Use appropriate markings / a scale and measure the fractional setting

Limitations	Improvements
Difficulty locating the exact centre or exact point from which a measurement should be taken	Use a fixed [ruler / grid] to locate the centre and a [pointer / mark / stop] to indicate the exact measurement position. Measure from both sides and **[average / account for the radius]** to determine the *centre*.
Taking the measurement disturbs or changes the apparatus The specimen deforms due to loading, handling or measurement The physical properties of the material change after repeated use or loading	Use a non-contact measuring method such as a travelling microscope / video with ruler to scale Reduce time under load by take readings quickly and allow recovery Use a new identical specimen every time an experiment is done
Instrument zero, calibration or stated-value does not match actual value.	Zero/check/calibrate the instrument or stated value Measure the actual value and account for it in calculation
Two supposedly identical components have different physical properties.	Measure the relevant property of **each** specimen/component and use its actual value. Use a new identical specimen every time an experiment is done
Thickness of a string, mark, pointer or contact makes the exact measurement point unknown	Use a thinner/narrower reference component — 16 Measure and correct for the finite thickness/width — 4
Glass may affect magnetic force / effect of surrounding magnetic materials (e.g. G clamp).	Use **non-magnetic supports/clamps**, move magnetic materials away, or test whether the separating/support material itself affects the result.

Stability

Limitations	Improvements
Difficulty setting or maintaining a component vertically / horizontally Hand-held measuring instrument cannot be kept sufficiently steady A component or support moves when it should remain fixed Components cannot be aligned accurately with the required direction or line One operator must control/release/hold the apparatus while simultaneously observing or taking another reading	Clamp ruler / protractor / [component] (vertically / horizontally) Use a set square / plumb line to ensure the component is vertical Use a spirit level to ensure the component is horizontal
A component slips from its intended position during the experiment An attachment or connection becomes detached during the experiment	Adhere with glue/tape/putty/Velcro or tie the component together Use a rough surface /groove / notch to prevent slippage

Motion:

Limitations	Improvements
One operator must control/release/hold the apparatus while simultaneously observing or taking another reading	Use another stand/ clamp / support to hold the [component] in position.
A quantity that should remain constant changes during the experiment	Fix/mark the controlled setting so it stays the same Use finer/continuous adjustment of the controlled variable

Limitations	Improvements
Friction at a pivot, pulley or contact surface affects the motion or force	Reduce friction with **lubricant**/smoother contact
Applied force cannot be kept in the required direction, position or magnitude	Use a pulley/mass/motor arrangement to apply a steady controlled force
The contact surface is not uniform, so friction/contact conditions vary with position.	Mark and use the **same section** of the surface for each trial, or use a more uniform surface

Pendulums:

Limitations	Improvements
Oscillation in more than one direction The object does not follow the intended path of motion Unintended paths of motion causing misaligned or inconsistent collisions. Environmental conditions disturb the apparatus or affect the reading	Use guides to restrict direction of oscillation. Change dimensions/spacing / alignment before release to prevent unintended contact Reduce draughts (the flow of cool air) disturbing the experiment
Release position cannot be reproduced consistently	Use a **mechanical hand** to release the object. Determine actual release position from video recording. Mark/fix the same release position
Oscillation pattern dies away quickly, too few usable oscillations	Reduce damping/thicken material or material so oscillations persist longer Use smoother sand board

Light:

Limitations	Improvements
Difficulty deciding when an image is in sharp focus / when the sharpest image has been obtained.	Darken the room Use object with sharper edges to focus easier e.g. cross-hairs/filament lamp/smaller LEDs.

Liquids and Air

Limitations	Improvements
The required scale, mark, surface or object is difficult to observe clearly Optical and liquid-surface effects on readings	Mark the liquid surface more clearly (with marker) Use coloured water /screen behind the objects / contrasting colours (also some other apparatus too)
Liquid remains in the apparatus during transfer, changing the actual quantity transferred Residue from a previous trial dilutes the next sample	Rinse / Dry components after using Measure the new liquid after every transfer
Wetting or absorption changes the dimensions or properties of the material	Use waterproof/non-absorbent material
Air bubbles or trapped air affect the effective volume or behaviour of the system	Remove/flush trapped air before measurement
Leakage causes the amount of liquid or gas in the apparatus to change	Seal/contain the fluid to stop leakage

Electricity

Limitations	Improvements
Electrical reading stability, meter response or circuit faults Electrical contact resistance or connection resistance affects the measured values	Clean/tighten/solder electrical contacts (with alcohol) Use a power supply (instead of battery) Allow reading to reach steady value. Measure lead resistance and subtract from R

Thermal

Limitations	Improvements
Heat is transferred to the surroundings or apparatus instead of remaining in the system	**Use insulation/lagging around the container**, e.g. **polystyrene** Use lid on container.
Temperature is spatially non-uniform, changes with time, or the sensor does not track the true temperature promptly	Stir/use a thermostatically controlled water bath Measure temperature at the required time / wait for equilibrium
Temperature continues to rise after switching off	Wait until temperature reaches a maximum before reading
Resistor/bulb of temperature-dependent part is not completely immersed	Change the geometry/container so the temperature-dependent part is **fully immersed**

Waves:

Limitations	Improvements
Difficult to produce exact [string] pattern required Difficult to see exact [string] pattern required	Use screen / background behind the objects / contrasting colours alter excitation position/load so the required node occurs photo/video to assess the pattern; (see difficult to judge)
String pattern dies away quickly	alter method to maintain continuous oscillation move excitation point away from node to create nodes use heavier mass to create nodes

“Difficulty / error / uncertainty + what physically causes it + what quantity or behaviour is affected.”

NO: “reference point problem”

> **Difficulty identifying the exact centre, endpoint or reference position**

NO: **“**small measurement”

> **Measured value or change is small, giving a large percentage uncertainty**

NO: “release problem”

> **Release conditions cannot be reproduced consistently**